Yuxiao Qu

cohenqu.github.io
Google Scholar: https://scholar.google.com/citations?user=X5SFuyYAAAAJ&hl=en

Education

Carnegie Mellon University — **Pittsburgh, PA**
PhD in Machine Learning, 4.22/4.33 — *May 2024 - Present*
Advisor: Prof. Aviral Kumar & Prof. Ruslan Salakhutdinov

Carnegie Mellon University — **Pittsburgh, PA**
Master of Science in Machine Learning , 4.28/4.33 — *Aug 2023 - Present*
Advisor: Prof. Aviral Kumar

University of Wisconsin - Madison — **Madison, WI**
Bachelor of Science in Computer Science and Mathematics, 4.00/4.00 — *Sep 2020 – Dec 2022*

The Chinese University of Hong Kong — **Sha Tin, Hong Kong**
Major: Computer Science — *Sep 2017 – May 2020*

Awards and Fellowships

- 2025 Amazon PhD Fellowship In AI — 2025
- 2025 Qualcomm Innovation Fellowship (*Finalist*) — 2025
- 2021 ICPC North Central NA Regional Contest (*Rank 19th*) — 2021
- Professor Charles K. Kao Research Exchange Scholarship — 2021
- Best Project Award in Undergraduate Summer Research Internship — 2019

Publications

(* indicates Equal Contribution)

1. **Learning to Discover Abstractions for LLM Reasoning**
   Yuxiao Qu*, Anikait Singh*, Yoonho Lee*, Amrith Setlur, Ruslan Salakhutdinov, Chelsea Finn, Aviral Kumar
   Spotlight at Workshop on Efficient Systems for Foundation Models, ICML 2025
   Oral at RAM 2: Reasoning, Attention & Memory – 10 Years On, COLM 2025
2. **Optimizing Test-Time Compute via Meta Reinforcement Fine-Tuning**
   Yuxiao Qu*, Matthew Y.R. Yang*, Amrith Setlur, Lewis Tunstall, Edward Emanuel Beeching, Ruslan Salakhutdinov, Aviral Kumar
   Oral at Workshop on Foundation Models in the Wild, ICLR 2025
   International Conference on Machine Learning (ICML), 2025
3. **Harnessing Webpage UIs for Text-Rich Visual Understanding**
   Junpeng Liu, Tianyue Ou*, Yifan Song*, **Yuxiao Qu***, Wai Lam, Chenyan Xiong, Wenhu Chen, Graham Neubig, Xiang Yue
   International Conference on Learning Representations(ICLR), 2025
4. **Recursive Introspection: Teaching Language Model Agents How to Self-Improve**

Yuxiao Qu, Tianjun Zhang, Naman Garg, Aviral Kumar
Oral at Workshop on Large Language Models and Cognition, ICML 2024
Advances in Neural Information Processing Systems (NeurIPS), 2024

5. **Guided Data Augmentation for Offline Reinforcement Learning and Imitation Learning**
   Nicholas E. Corrado, **Yuxiao Qu**, John U. Balis, Adam Labiosa, Josiah P. Hanna
   Reinforcement Learning Conference (RLC), 2024
6. **FLEE-GNN: A Federated Learning System for Edge-Enhanced Graph Neural Network in Analyzing Geospatial Resilience of Multicommodity Food Flows**
   Yuxiao Qu, Jinmeng Rao, Song Gao, Qianheng Zhang, Wei-Lun Chao, Yu Su, Michelle Miller, Alfonso Morales, Patrick Huber
   ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery, 2023
7. **Simulation-Acquired Latent Action Spaces for Dynamics Generalization**
   Nicholas E. Corrado, **Yuxiao Qu**, Josiah P. Hanna
   Proceedings of Machine Learning Research (PMLR), 2022
8. **Polite or Direct? Conversation Design of a Smart Display for Older Adults Based on Politeness Theory**
   Yaxin Hu, **Yuxiao Qu**, Adam Maus, Bilge Mutlu
   ACM CHI Conference on Human Factors in Computing Systems (CHI), 2022
9. **TreeNet: Deep Point Cloud Embedding for Routing Tree Construction**
   Wei Li, **Yuxiao Qu**, Gengjie Chen, Yuzhe Ma, Bei Yu
   Best Paper Award at ACM Asian and South Pacific Design Automation Conference (ASPDAC), 2021

Research Blog Posts

- **Optimizing LLM Test-Time Compute Involves Solving a Meta-RL Problem**, *CMU ML Blog*

Experiences

Teaching Assistant

- COMP SCI 577: Introduction to Algorithms (*Honor Session*) — Fall 2022
- COMP SCI 537: Introduction to Operating Systems — Fall 2022

Employment

Morgridge Institute for Research — **Madison, WI**
Research Software Engineer — *Apr 2023 – Aug 2023*

Professional Service

- Reviewer for ICML (2024), NeurIPS (2024-2025), AISTATS (2025), ICLR (2025)